

Breadth - First Search [BFS]

- * BFS Explores the graph level by level.
- * It starts from a given source node.
- * Visits all the neighbours of the current node before moving to the next level neighbours.

Data Structures: 1) Queue: used FIFo for exploration.

2) Visited Array/Set: Visited nodes to avoid cycles and repeated work.

Time Complexity: $O(N+E) \rightarrow N = \text{Nodes} / E = \text{Edges}$
Nodes & Edges are explored once

Depth - First Search.

- * DFS Explores down a branch as possible
- * Marks the node visited.
- * Starts from the node and explores one branch fully before moving to another

Data Structures: Stack: used LIFO Exploration
Visited Array/List: Visited nodes are stored.

Time Complexity: $O(N+E) \rightarrow N = \text{Nodes} / E = \text{Edges}$
Each node & Edges are visited once